Math 510 **Fall 2026**
Homework 4 **Drew Armstrong**

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1. Evaluating Polynomials at Matrices. Given a polynomial $f(x) = \alpha_0 + \alpha_1 x + \cdots + \alpha_d x^d \in \mathbb{C}[x]$ and a square matrix $A \in \mathbb{C}^{n\times n}$, we define the matrix

$$f(A) := \alpha_0 I_n + \alpha_1 A + \alpha_2 A^2 + \cdots + \alpha_d A^d \in \mathbb{C}^{n\times n}.$$

Note that for diagonal matrices we have

$$f\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = \begin{pmatrix} f(\lambda_1) & & \\ & \ddots & \\ & & f(\lambda_n) \end{pmatrix}.$$

(a) If $f(A) = O$ is the zero matrix, prove that every eigenvalue $\lambda \in \mathbb{C}$ of A satisfies $f(\lambda) = 0$.
(b) If $A = CBC^{-1}$, prove that $f(A) = Cf(B)C^{-1}$.

Both properties also hold for power series $f(x)$, assuming convergence.

2. Linear Dynamical Systems. Let $\mathbf{x}(t) = (x_1(t), \ldots, x_n(t))$ denote a moving point in $\mathbb{R}^n$, with velocity vector $\mathbf{x}'(t) = (x_1'(t), \ldots, x_n'(t))$. Consider a real matrix $A \in \mathbb{R}^{n\times n}$ and suppose that we have $\mathbf{x}'(t) = A\mathbf{x}(t)$ for all t. The general solution is given by

$$\mathbf{x}(t) = \exp(tA)\mathbf{x}(0).$$

Use this theorem to solve the following system:

$$\begin{cases} x'(t) &= 0 &- y(t), \\ y'(t) &= x(t) &+ 0. \end{cases}$$

[Hint: Write the system in the form $\mathbf{x}'(t) = A\mathbf{x}(t)$. Explicitly diagonalize the matrix A over $\mathbb{C}$ and then use the fact that $A = X\Lambda X^{-1}$ implies $\exp(tA) = X\exp(t\Lambda)X^{-1}$ from 1(b).]

3. Idempotent Matrices. Let $P \in \mathbb{C}^{n\times n}$ satisfy $P^2 = P$.

(a) Show that the eigenvalues of P are in the set $\{0, 1\}$. [Hint: Use 1(a).]
(b) Let E_0 and E_1 be the 0- and 1-eigenspaces of P.[1] Prove that

$$\mathbb{C}^n = E_0 \oplus E_1.$$

[Hint: For any $\mathbf{x} \in \mathbb{C}^n$ we can write $\mathbf{x} = (\mathbf{x} - P\mathbf{x}) + P\mathbf{x}$. Show that $\mathbf{x} - P\mathbf{x} \in E_0$ and $P\mathbf{x} \in E_1$.]
(c) Suppose $\dim E_1 = r \geq 1$. Let $\mathbf{x}_1, \ldots, \mathbf{x}_r$ be a basis for E_1 and let $\mathbf{x}_{r+1}, \ldots, \mathbf{x}_n$ be a basis for E_0, so that $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{C}^n$ is a basis for $\mathbb{C}^n$. Let $X \in \mathbb{C}^{n\times n}$ be the matrix with columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$, which is invertible. Show that

$$P = X\left(\begin{array}{c|c} I_r & O_{r,n-r} \\ \hline O_{n-r,r} & O_{n-r,n-r} \end{array}\right)X^{-1}.$$

(d) Use part (c) to prove that $P = AB^*$ for some matrices $A, B \in \mathbb{C}^{n\times r}$ satisfying $B^*A = I_r$. [Hint: Let A be the first r columns of X and let B^* be the first r rows of X^{-1}.]

4. Independence of Eigenvectors. Let $A \in \mathbb{C}^{n\times n}$ and let $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{C}^n$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$. By distinct, we mean that $i \neq j$ implies $\lambda_i \neq \lambda_j$.

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[1] It is possible that one of these eigenspaces is trivial.

(a) For each index i, consider the polynomial

$$p_i(x) = \prod_{j \neq i} \frac{x - \lambda_j}{\lambda_i - \lambda_j} \in \mathbb{C}[x].$$

Note that $p_i(\lambda_j) = \delta_{ij}$. Use this to show that $p_i(A)\mathbf{x}_j = \delta_{ij}\mathbf{x}_j$.

(b) Use part (a) to prove that $\mathbf{x}_1, \ldots, \mathbf{x}_k$ is an independent set. [Hint: Suppose $\sum_j \alpha_j \mathbf{x}_j = \mathbf{0}$. Multiply both sides by the matrix $p_i(A)$.]

5. Normal Operators. Let V be a Hilbert space and let $A : V \to V$ be a continuous operator satisfying $A^*A = AA^*$. (You can think of $V = \mathbb{C}^n$, in which case A^* is the conjugate transpose matrix.)

(a) Prove that $\langle A\mathbf{x}, A\mathbf{y}\rangle = \langle A^*\mathbf{x}, A^*\mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y} \in V$.

(b) Show that $A\mathbf{x} = \mathbf{0}$ if and only if $A^*\mathbf{x} = \mathbf{0}$. [Hint: Taking $\mathbf{y} = \mathbf{x}$ in part (a) gives $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$.]

(c) Show that $A\mathbf{x} = \lambda\mathbf{x}$ if and only if $A^*\mathbf{x} = \lambda^*\mathbf{x}$, hence A and A^* have the same eigenvectors. [Hint: Consider the operator $B = A - \lambda I$. Show that $B^*B = BB^*$ and then use part (b).]

(d) If $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$ with $\lambda \neq \mu$, prove that $\langle \mathbf{x}, \mathbf{y}\rangle = 0$. [Hint: Use part (c) to show that $\lambda\langle \mathbf{x}, \mathbf{y}\rangle = \mu\langle \mathbf{x}, \mathbf{y}\rangle$.]

There will be one or two more problems on the topics:

- Total least squares.
- Perron-Frobenius and Markov chains.